2

Solutions of Non-Linear Equations in One Variable

Chapter Learning Outcomes

CLO

2

- Understand the root finding problem.
- List some methods which are used to solve nonlinear equations in one variable.
- Distinguish between bracketing and open methods.
- Know and understand the intermediate value theorem.
- Apply the bisection method to solve nonlinear equations in one variable.
- Find the number of iterations needed to solve nonlinear equations in one variable using the bisection Method.
- Apply Newton-Raphson's method to solve nonlinear equations in one variable.
- Apply Secant method to solve nonlinear equations in one variable.
- Distinguish between converging and diverging.

Root Finding Problems

Many problems in Science and Engineering are expressed as:

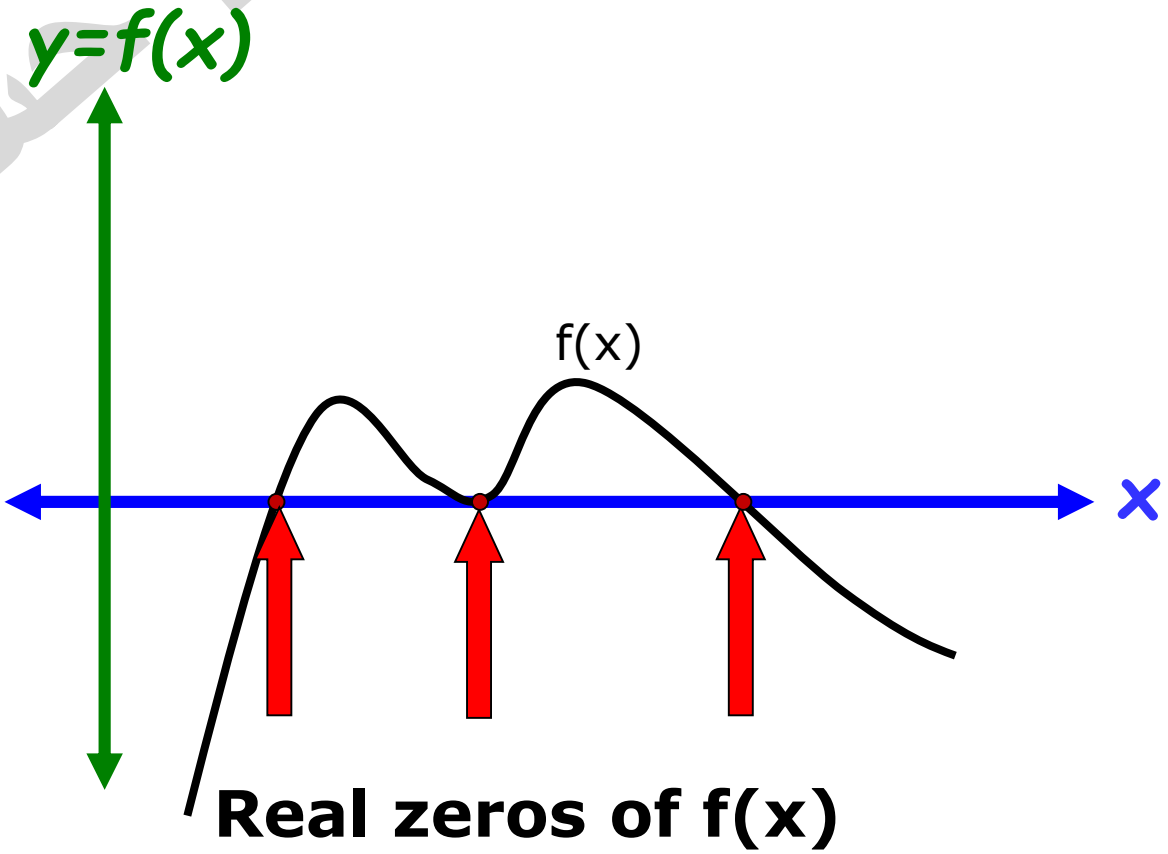
Given a continuous function $f(x)$,
find the value r such that $f(r) = 0$

These problems are called root finding problems.

Zeros of a Function

Let $f(x)$ be a real-valued function of a real variable. Any number r for which $f(r)=0$ is called a zero of the function.

The real zeros of a function $f(x)$ are the values of x at which the graph of the function crosses (or touches) the x-axis.



Zeros of a Function

Example

Find the zeros for the function $f(x) = (x-2)(x-3)$.

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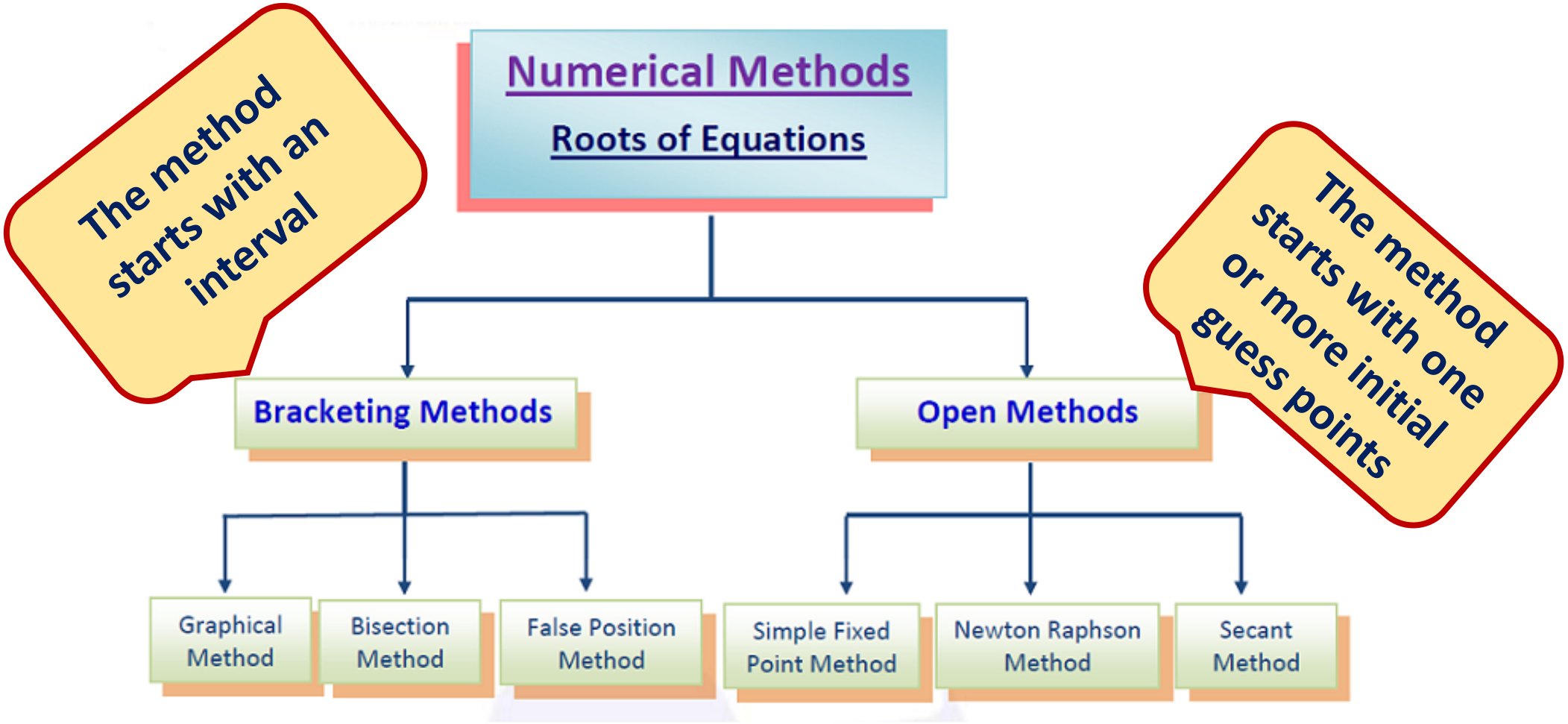
Solutions of Non-Linear Equations in One Variable

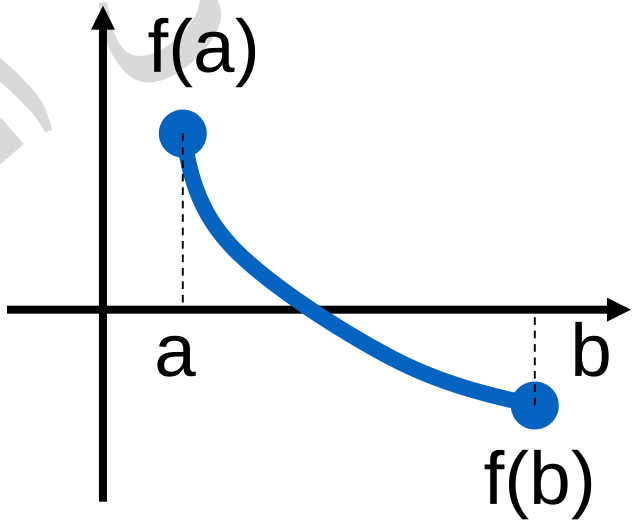
Many methods are available to solve nonlinear equations:



Bisection Method

Intermediate Value Theorem

Definition 1.1 Let f(x) be a continuous function defined on [a,b]; such that $f(a) \times f(b) < 0$ (f(a) and f(b) are opposite signs - different signs). Then there is at least one root in the interval [a,b].



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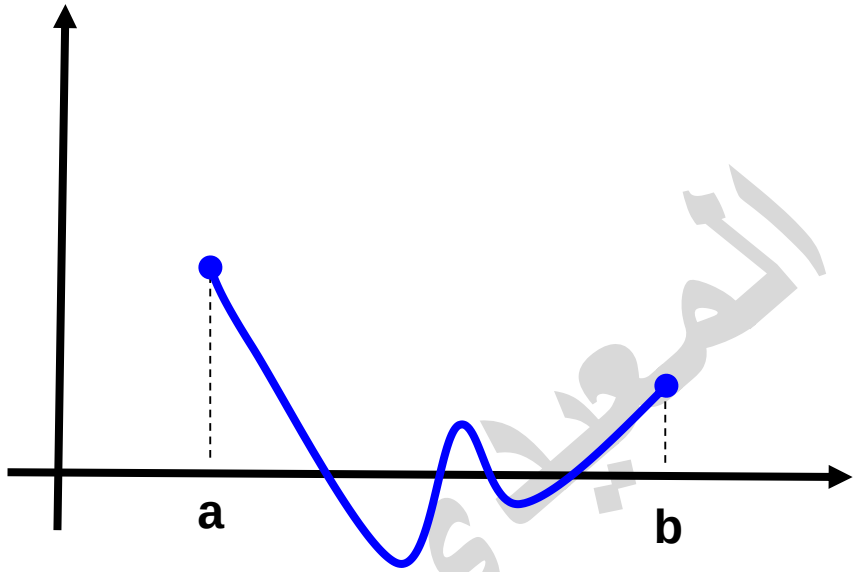
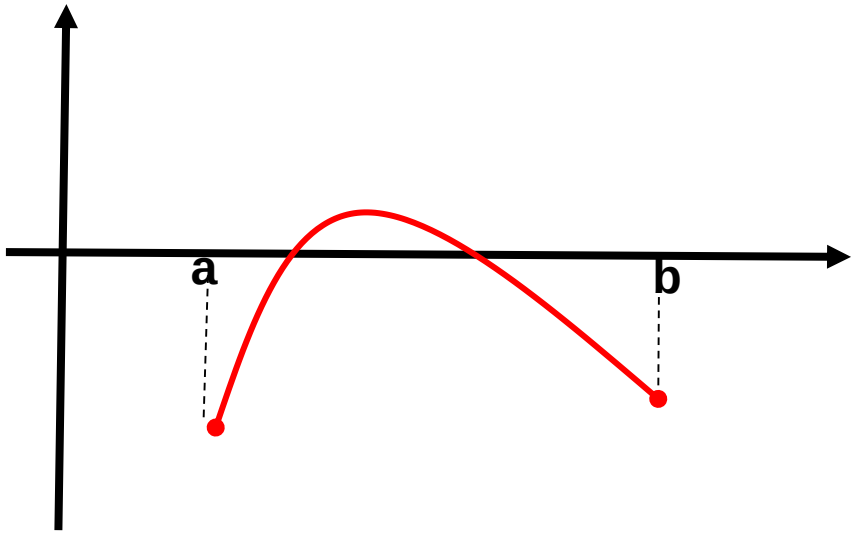
Intermediate Value Theorem

If f(a) and f(b) have the same sign, the function may have an even number of real zeros or no real zeros in the interval [a, b].

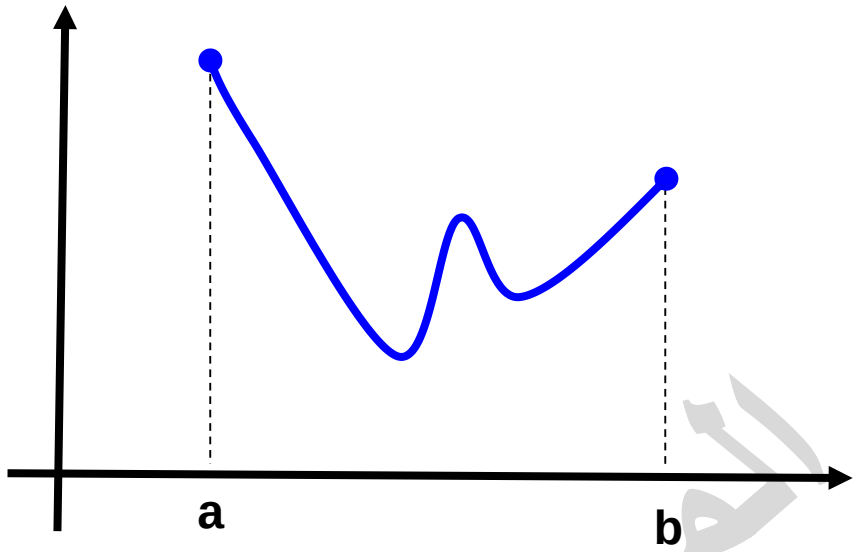


The function has four real zeros

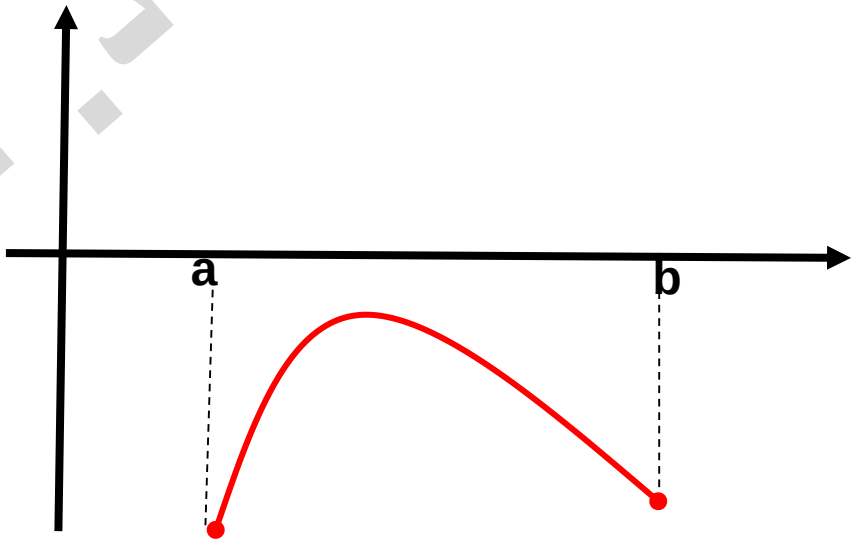


The function has two real zeros

Intermediate Value Theorem



The function has no real zeros



The function has no real zeros

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Intermediate Value Theorem

Example

Let $f(x) = x^5 + x^4 - 3$, prove that f has a root x in the interval $[1,2]$.

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Bisection Method

- The **Bisection method** is one of the simplest methods to find a zero of a nonlinear function.
- It is also called **interval halving** method.
- To use the Bisection method, one needs an initial interval [a, b] that is known to contain a zero of the function.
- The method systematically reduces the interval. It does this by dividing the interval into two equal parts, performs a simple test and based on the result of the test, half of the interval is thrown away.
- The procedure is repeated until the desired interval size is obtained.

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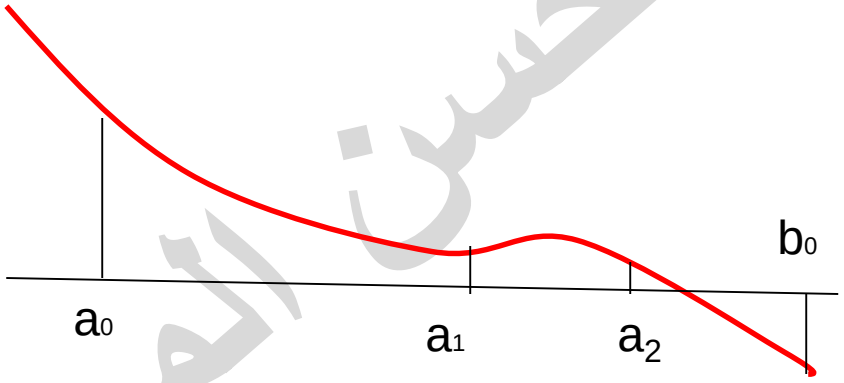
Bisection Method



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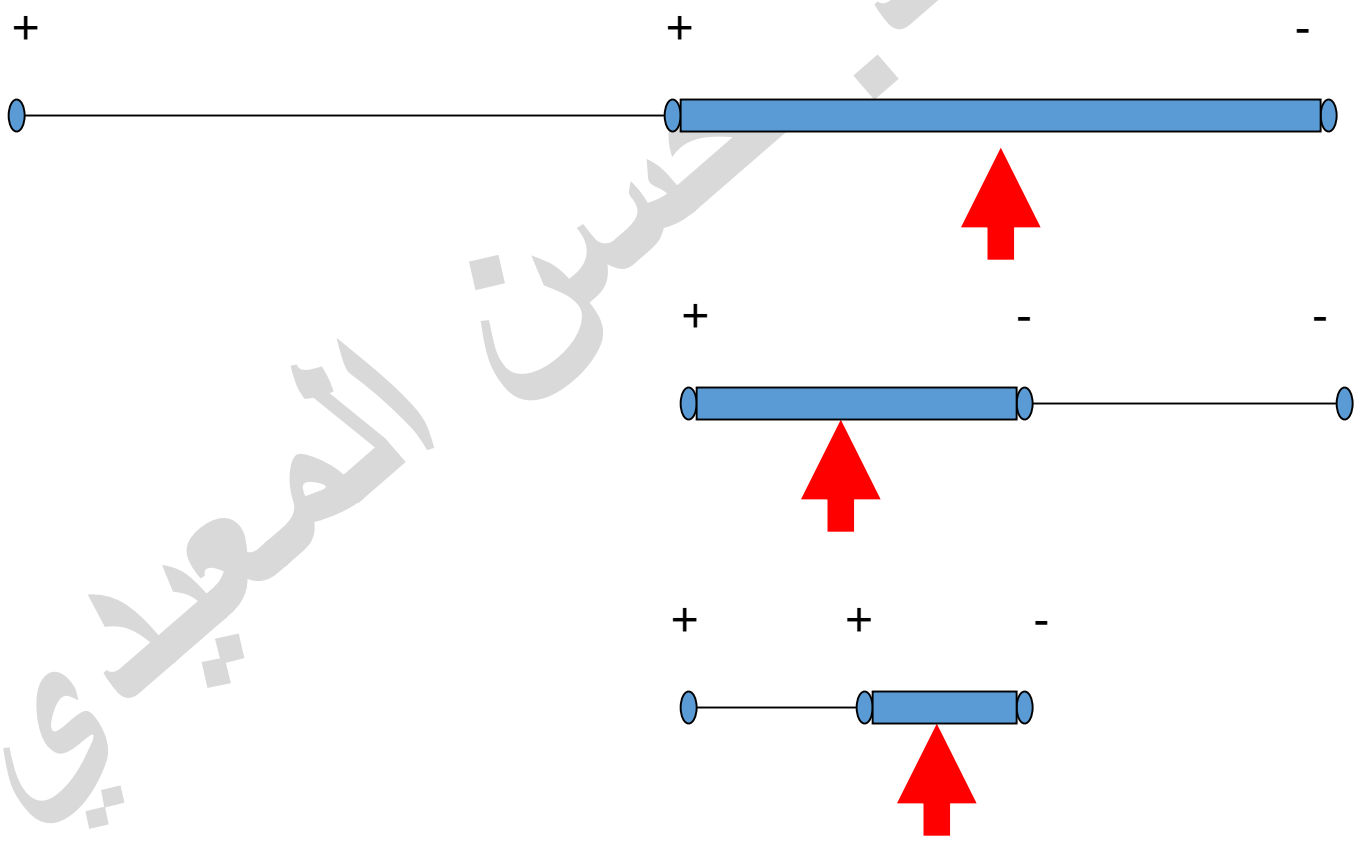
Bisection Method



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Bisection Method

EXAMPLE

Can you use Bisection method to find a zero of :

$f(x) = x^3 - 3x + 1$ in the interval $[0,2]$?

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Bisection Method

EXAMPLE

Can you use Bisection method to find a zero of :

$f(x) = x^3 - 3x + 1$ in the interval $[0,1]$?

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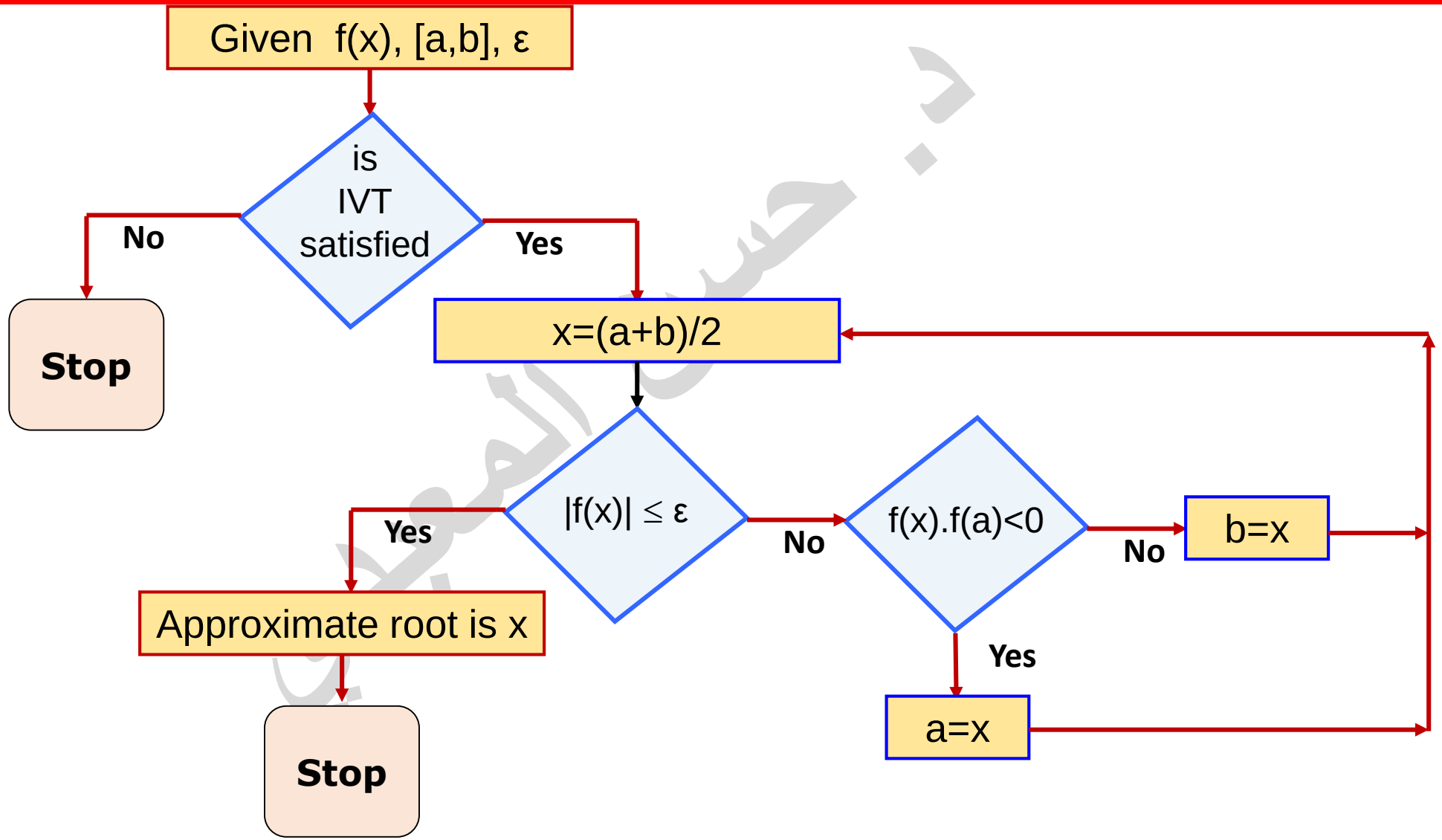
Bisection Method Flowchart



Bisection Method Algorithm

Require: f(x), interval [a,b], tolerance (ϵ), maximum number of iterations (n).

```
if (f(a) × f(b) < 0) then
    Print "No Root in interval [a,b]"
    Exit algorithm.
end if
Set i=1
while i ≤ n do
    Compute x = (a + b)/2
    if (|f(x)| < ε) then
        Print x is the approximate root of f(x)
        Exit algorithm
    else if (f(x) × f(a) < 0) then
        b=x;
    else
        a=x;
    end if
    Increment i by 1
end while
Print "The method failed after n iterations"
```

Stopping Criteria

Three common stopping criteria

1. **Stop after a fixed number of iterations (n).**
2. **The functional value f(x) ≤ the tolerance (ε) … |f(x)| ≤ ε .**
3. **The length of the interval after k iterations < the tolerance** $\frac{b-a}{2^k} \leq \varepsilon$

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Bisection Method

EXAMPLE

Use Bisection method to determine a solution accurate to within 10^{-4} for $f(x) = x - 3 = 0$, in the interval [1.5, 3.5].

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Bisection Method

EXAMPLE

Use Bisection method to determine a solution accurate to within 10^{-3} for $f(x) = e^x - x - 2 = 0$, in the interval [-2.4, -1.6].

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Bisection Method

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Convergence Analysis

Given a, b, and ε.

How many iterations are needed to apply the bisection method.

$$n \geq \frac{\log(b-a)-\log(\varepsilon)}{\log(2)}$$

Convergence Analysis

EXAMPLE

Find the number of iterations needed to achieve an approximation with accuracy 10^{-3} to the solution of f(x)=0 lying in the interval [2.5, 4] using Bisection Method.

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Convergence Analysis

EXAMPLE

Find the number of iterations needed to achieve an approximation with accuracy 10^{-1} to the solution of f(x)=0 lying in the interval [0, 1] using Bisection Method.

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Quiz

Use Bisection method to determine a solution accurate to within 10^{-4} for $f(x) = x^3 + 4x^2 - 10 = 0$, in the interval [1, 2].

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Quiz

Use Bisection method to determine a solution accurate to within 10^{-3} for $f(x) = 30e^{-x} + \ln(x+1) - 3 = 0$, in the interval [0, 5].

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Quiz

Use Bisection method to determine a solution accurate to within 10^{-3} for $f(x) = 3x + \sin x - e^x = 0$, in the interval [0, 0.5].

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Newton Raphson Method

Newton's Method

- Newton Raphson method is one of the most powerful and well-known numerical methods for solving a root-finding problem.

- It's an **open method**.

- Given an initial guess of the root x_0, Newton-Raphson method uses information about the function and its derivative at that point to find a better guess of the root.

- **Assumptions:**
  - $f(x)$ is continuous and the first derivative is known.
  - An initial guess x_0 such that $f'(x_0) \neq 0$ is given.

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Newton's Formula

Given:

1. f(x), f'(x) where $f'(x_0) \neq 0$.
2. An initial guess x_0.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

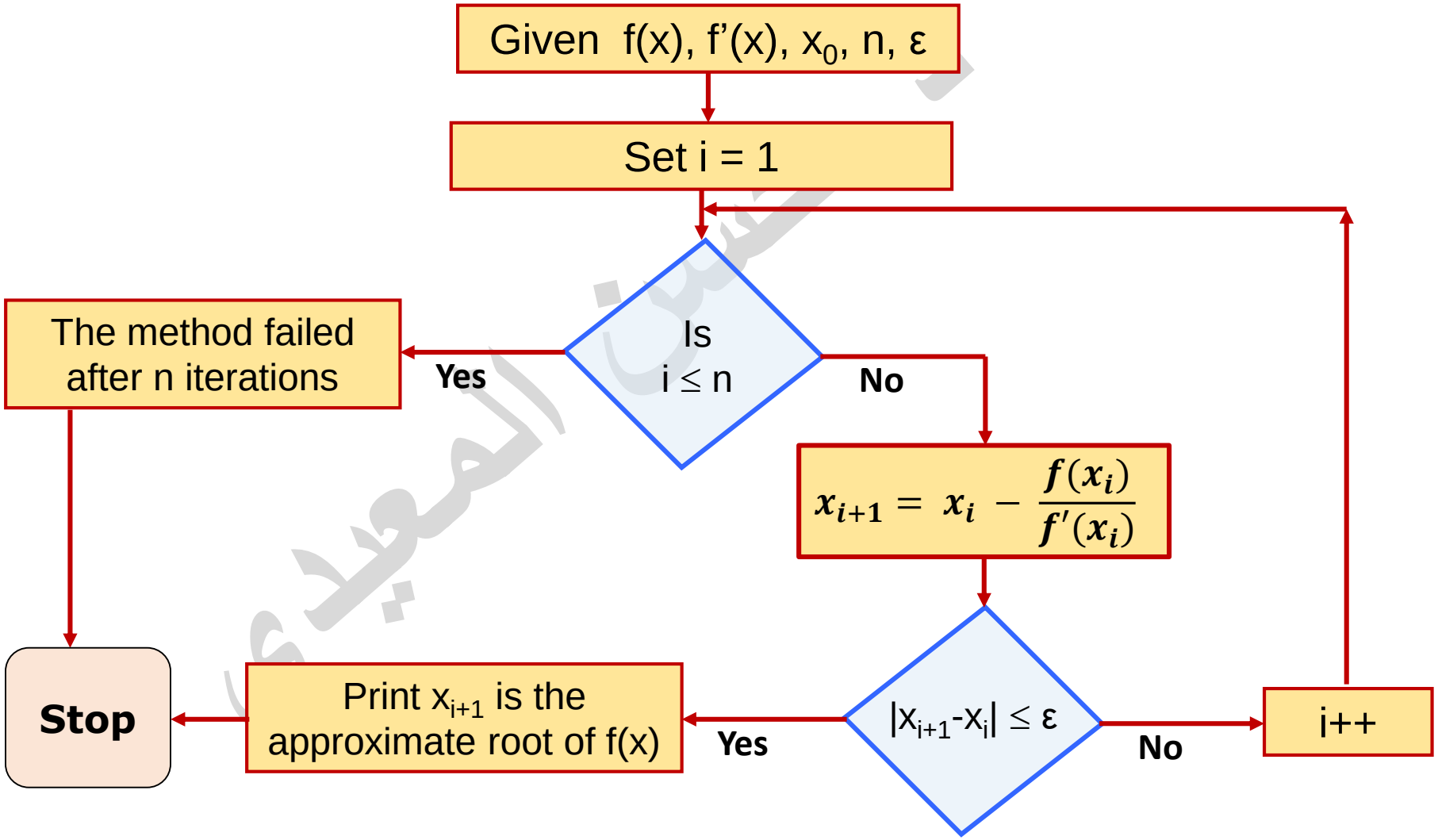
$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)}$$

$$\vdots$$

Newton's Formula

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Newton's Method Flowchart



Newton's Method Algorithm

Require: f(x), f'(x), tolerance (ϵ), initial value (x_1), maximum number of iterations (n).

Set i = 1.
while $i \leq n$ **do**
    Compute $x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}$.
    if ($|x_{i+1} - x_i| \leq \epsilon$) **then**
        Print x_{i+1} is the approximate root of f(x).
    end if
    Increment i by 1.
end while
Print "The method failed after n iterations".

Some Derivative Rules

Powers	$f(x) = x^n$	$f'(x) = nx^{n-1}$
Sine	$f(x) = \sin x$	$f'(x) = \cos x$
Cosine	$f(x) = \cos x$	$f'(x) = -\sin x$
Tangent	$f(x) = \tan x$	$f'(x) = \sec^2 x$
Exponential (base e)	$f(x) = e^x$	$f'(x) = e^x$
Exponential (base a)	$f(x) = a^x$	$f'(x) = a^x \ln a$
Natural Logarithm	$f(x) = \ln x$	$f'(x) = 1 / x$

Newton's Method

EXAMPLE

Use Newton's method to determine a solution accurate to within 10^{-2} for $f(x) = x^3 - x + 1 = 0$, assuming $x_0 = -1$.

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Newton's Method

EXAMPLE

Use Newton's method to determine a solution accurate to within 10^{-2} for $f(x) = x^3 - x + 1 = 0$, assuming $x_0 = -1$.

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Newton's Method

EXAMPLE

Use Newton's method to determine a solution accurate to within 10^{-3} for $f(x) = -x^3 - \cos(x) = 0$, assuming $x_0 = -1$.

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Quiz

Use Newton's method to determine a solution accurate to within 10^{-3} for $f(x) = \cos(x) - x = 0$, in the interval $[0, \pi/2]$.

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Quiz

Use Newton's method to approximate the value $\sqrt[3]{30}$ accurate to within 10^{-3}, assuming $x_0 = 3$.

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Secant Method

Secant Method

- What if we do not know the derivative of *f(x)*?
- Secant is an open method.
- Given an initial two guess points: x_0, and x_1.

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Secant Formula

Given:

1. f(x).
2. Two initial guess points x_0, x_1.

$$x_2 = x_1 - f(x_1)\frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$x_3 = x_2 - f(x_2)\frac{x_2 - x_1}{f(x_2) - f(x_1)}$$

.

.

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Secant Formula

$$x_{i+1} = x_i - f(x_i)\frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})}$$

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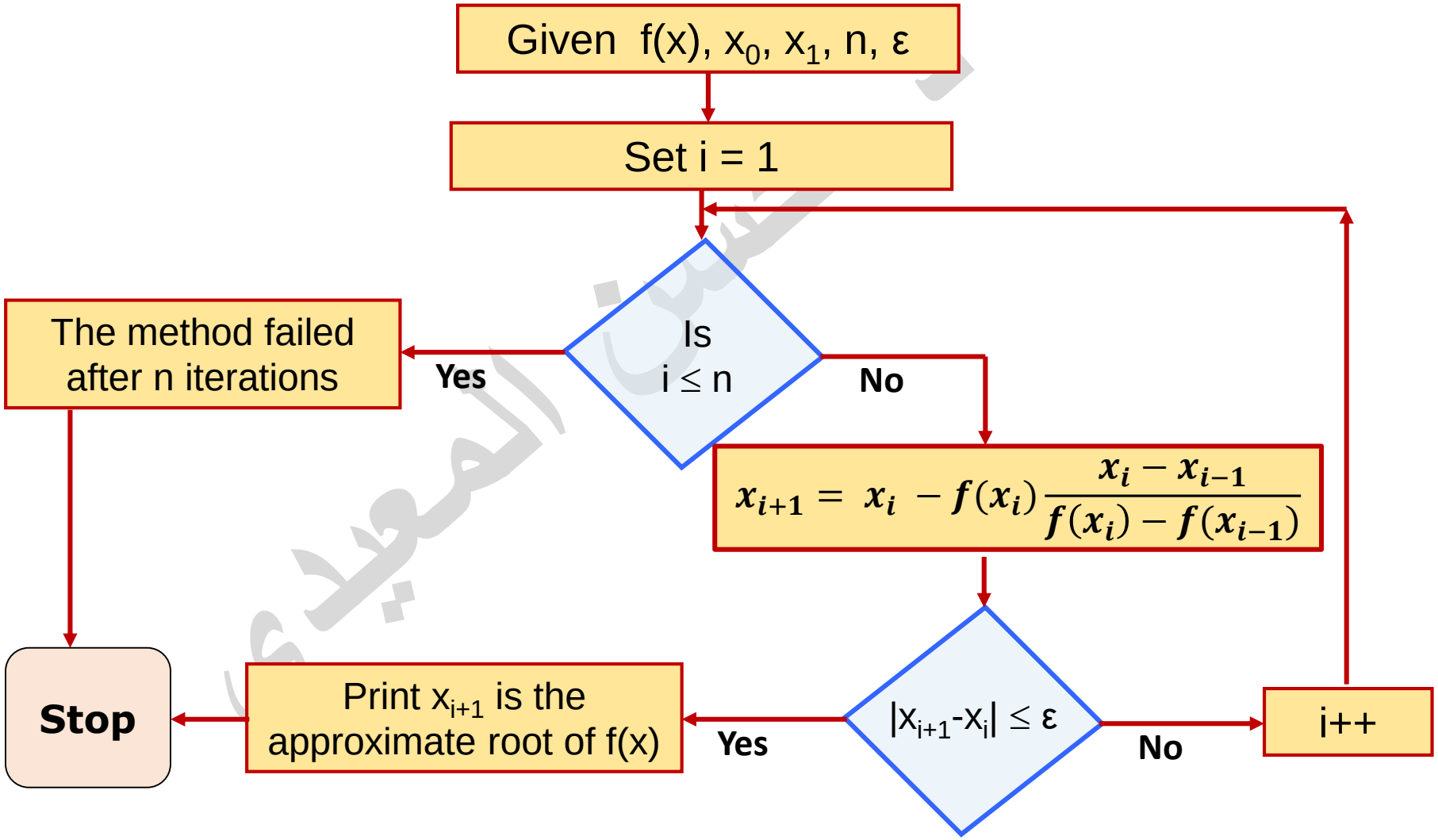
Secant Method Flowchart



Secant Method Algorithm

Require: f(x), tolerance (ϵ), two initial values (x_0, x_1), maximum number of iterations (n).

Set i = 2.
while $i \leq n$ **do**
    Compute $x_{i+1} = x_i - f(x_i)\dfrac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})}$.
    if ($|x_{i+1} - x_i| \leq \epsilon$) **then**
        Print x_{i+1} is the approximate root of f(x).
    end if
    Increment i by 1.
end while
Print "The method failed after n iterations".

Secant Method

EXAMPLE

Use Secant method to determine a solution accurate to within 10^{-3} for $f(x) = x^5 + x^3 + 3 = 0$, assuming $x_0 = -1$ and $x_1 = -1.1$.

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Secant Method

EXAMPLE

Use Secant method to determine a solution accurate to within 10^{-3} for $f(x) = -x^3 - \cos(x) = 0$, assuming $x_0 = -0.9$ and $x_1 = -1$.

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Quiz

Use Secant method to approximate the value $\sqrt[3]{30}$ accurate to within 10^{-3}, assuming $x_0 = 3.0$ and $x_1 = 3.1$.

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Check - Chapter Learning Outcomes

CLO

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End of Chapter

2